

Notes on replication of Ajzenman

System 1

- OS: "openSUSE Leap 15.6"
- Processor: AMD Ryzen 9 3900X 12-Core Processor, 24 cores
- Memory available: 125GB memory
- Docker version 27.5.1-ce, build 4c9b3b011ae4
- stata version 18-mp-i (Docker image dataeditors/stata18-mp-i:2024-12-18)

System 2

System 3

Bugs

- Variable name: `ren ientidad ent` was recoded as `ren entidad ent` (<https://github.com/larsvilhuber/replication-ajzenman-power-of-example/commit/95ea3eee0cec1ffd50ac6bf78180b66c849703a0>)
- Filename: one of 13 database files (`Data/Bases Madre/Homicides/DEFUN13.DBF`) has a different case, on a case-sensitive file system. All others are lower-case `dbf`. Manually modified the filename to be lower-case. (https://github.com/larsvilhuber/replication-ajzenman-power-of-example/blob/main/Corruption_Final/logs/logfile_30_Apr_2025-13_37_20-statauser.log#L811)
- Bug related to Stata version: `option contents()` not allowed since Stata 17; see [help table for updated syntax](#)

Structure

There are 143 `areg` regressions. The tables are not cleanly output.

Runtime: 1 hour 9 minutes (until bug)

Table 4:

To identify all coefficients from the separate regressions, I used `grep` to parse the Stata log file captured for `Table4.txt`:

```
grep -n "inter" Table4.txt | grep -v areg | awk ' { print $1, $2, $4, $5 } '
```

The output is:

```

      Coefficient  std.err. t    P>|t|
25:  inter  .2588744 .0955364 2.71 0.008
87:  inter  .4995433 .1963678 2.54 0.013
219: inter  .5450504 .1686364 3.23 0.002
364: inter  .5299935 .1729945 3.06 0.003
618: inter  .6067147 .1414045 4.29 0.000 <== Column 5
834: inter  .1218765 .0556162 2.19 0.031
895: inter  .1802902 .0995852 1.81 0.074
1027: inter  .1839404 .0886252 2.08 0.041
1166: inter  .1975147 .0725312 2.72 0.008
1323: inter  .1856933 .0706579 2.63 0.010 <== Column 5
1538: inter  .1106731 .0370288 2.99 0.004
1599: inter  .2172922 .0713383 3.05 0.003
1731: inter  .2479733 .0547653 4.53 0.000
1870: inter  .2607372 .045871 5.68 0.000
2027: inter  .2590473 .0523933 4.94 0.000 <== Column 5
2243: inter  .0756975 .0211231 3.58 0.001
2304: inter  .1179639 .0411609 2.87 0.005
2436: inter  .1242549 .0370961 3.35 0.001
2575: inter  .1416991 .027053 5.24 0.000
2732: inter  .1398497 .0295395 4.73 0.000 <== Column 5

```

Reverse engineering variable names:

- **Uno** = "at least 1 incivic value"
- **Dos** = "at least 2 incivic values"
- **Tres** = "1st component PCA with 4 incivic values"
- **Count_Index** = "Count index (sum 5 incivic values)"

Robustness check 1

Checked variable definitions

- **Count_Index**:

```

** general index using the 5 vars
egen Count_Index=rowtotal(Indice_Tramposo Indice_Tramposo2 Confiable)
replace Count_Index = . if Indice_Tramposo ==. | Indice_Tramposo2 ==. |
Confiable ==.

```

- **Uno**:

```

Values.do:503:gen Uno = 0 if Count_Index <1
Values.do:504:replace Uno = 1 if Count_Index >=1

```

```
Values.do:505:replace Uno = . if Count_Index ==.
```

- Dos:

```
Values.do:509:gen Dos = 0 if Count_Index <2
Values.do:510:replace Dos = 1 if Count_Index >=2
Values.do:511:replace Dos = . if Count_Index ==.
```

- Tres:

```
pca Romper NotReturn Robar Tranza Confiable
predict pc_1
egen A = min(pc_1)
egen B=max(pc_1)
gen Dif=B-A
gen index=(pc_1-A)/Dif
ren index Tres
```

The variable of interest is `CorruptAfter(mtf)` and is the interaction between `Corrupt(mt)` (a dummy that scores 1 if the municipality *m* is corrupt during a particular period *t*) and `After(f)` (a dummy that takes on a value of 1 if the interview of the individual was performed before the month of February, when the reports were released, and 0 otherwise).

```
keep clave_mun year unautho Auditada Veces Corr_Acum Corrupt* una*
CorruptPast* C_menos1 C_menos2 C_1 C_2 lag* lead*
sort clave_mun Aud year
bysort clave_mun Aud : gen nro_auditoria = _n if Aud == 1
replace Auditada = 0 if Auditada ==.
replace nro = 0 if nro ==.
replace unau = 0 if unau ==.
...
gen Corrupt = 0
replace Corrupt =1 if unau >0
```

Test code:

```
include "config.do"
global cdname "$rootdir"

global cd_do "$cdname/Do"
global cd_data "$cdname/Data"
```

```

use "$cd_data/Longitudinal/Claves.dta", clear
** Delete obs that are not in the survey (but yes in the corruption dataset)
** merge with corruption again
merge 1:m clave_mun using
"$cd_data/Data_Corruption/TodosLosMuni_ConAudits_ConAlreadyAudited.dta"
keep if _merge !=2
desc
drop Corrupt
gen Corrupt = 0
replace Corrupt =1 if unau >0
tab Corrupt

```

Results

```
. tab Corrupt
```

Corrupt	Freq.	Percent	Cum.
0	1,370	85.31	85.31
1	236	14.69	100.00
Total	1,606	100.00	

Definitions: "Count index": sum of five incivic answers; "At least 1": at least one incivic answer; "At least 2": at least two incivic answers; "PC": First component of a PCA (normalized to a 0–1 scale). The exact wording of the individual questions is as follows: (i) "The one who does not cheat, does not get ahead" (Completely Agree, Agree, Disagree, Completely Disagree); (ii) "Are you trustworthy?" (Completely Agree, Agree, Disagree, Completely Disagree); (iii) "Laws were made to be broken" (Completely Agree, Agree, Disagree, Completely Disagree); (iv) "How likely is it that you steal electricity from the public lines (illegally)?" (1 to 100); (v) "How likely is it that you return a wallet with 500 pesos in it?" (1 to 100). In brackets: the estimated coefficients divided by the standard deviation of each variable.

Coding

Romper is coded 1 if respondent agrees or completely agrees. Median = Completely disagree.

Trustworthiness **Confiable** is reverse coded: it is set to 1 if the respondent disagrees or completely disagrees. Median = Agree.

Steal electricity is coded 1 if the respondents answer is above the mean. Mean = 6.58%.

Return wallet is coded 1 if the respondents answer is above the mean. Mean = 64.20%. Self-reported approx. probability: 81.30%

* break the rules

```
gen Romper = 0 if co01 !=. & co01 !=8  
replace Romper =1 if inlist(co01,1,2)
```

* to get ahead in life need to cheat

```
gen Tranza = 0 if co03 !=. & co03 !=8  
replace Tranza =1 if inlist(co03,1,2)
```

* trustworthy

**

```
gen Confiable = 1 if co05 !=8 & co05!=.  
replace Confiable =0 if inlist(co05,1,2)
```

**

** steal electricity

```
su rg09_11 if rg09_11 >0  
return list  
gen Media_Robar = r(mean)  
egen p90 = pctl(rg09_11), p(90)
```

```
gen Robar = 0 if rg09_11 !=.  
replace Robar = 1 if rg09_11 >Media_Robar  
replace Robar = . if rg09_11==.
```

** not return wallet (with the opposite sign to make it compatible with the rest)

```
gen complemento = 100-rg12_11 if rg12_11 !=.
```

```
su complemento if complemento >0  
return list  
gen Media_NotReturn = r(mean)  
egen nrp90 = pctl(complemento), p(90)
```

```
gen NotReturn = 0 if complemento !=.  
replace NotReturn = 1 if complemento >Media_NotReturn  
replace NotReturn = . if complemento ==.
```

Changes these to be more extremely "poor judgement": limited to **completely agree** (or reverse coded **completely disagree**), and to the 90 percentiles.

Results:

```

            Coefficient  std.err.  t    P>|t|
25:  inter  -.0285057  .1090595 -0.26 0.795
87:  inter  .0817852  .1547545 0.53 0.599
219: inter  .0600518  .1328355 0.45 0.653
364: inter  -.0051433  .1228959 -0.04 0.967
618: inter  .0994415  .1170965 0.85 0.398      <== Column 5
834: inter  -.0390027  .0753067 -0.52 0.606
895: inter  .0107975  .1158709 0.09 0.926
1027: inter -.0100146  .1081729 -0.09 0.926
1166: inter  .0306619  .1096336 0.28 0.780
1323: inter -.0048173  .1015055 -0.05 0.962      <== Column 5
1538: inter  .0059874  .0325353 0.18 0.854
1599: inter  .0412236  .0484137 0.85 0.397
1731: inter  .0416201  .0429528 0.97 0.336
1870: inter  .0695075  .0332821 2.09 0.040
2027: inter  .0520482  .0291082 1.79 0.078      <== Column 5
2243: inter  .0414757  .0115817 3.58 0.001
2304: inter  .0687284  .0193136 3.56 0.001
2436: inter  .0613907  .0178548 3.44 0.001
2575: inter  .0658179  .0188798 3.49 0.001
2732: inter  .0599955  .0213624 2.81 0.006      <== Column 5

```

Strongly reduces magnitude (by about half), and significance of many of the coefficients (mostly the PCA remains significant, none of the others at conventional levels).

- Program: [Corruption_Final/Do/Values_robustness.do](#)
- Output: [Corruption_Final/Do/Values_robustness.log](#) and [Corruption_Final/Tables/Table4-robustness1.txt](#)

Robustness check 2

- Program: [Corruption_Final/Do/robustness_GAB_1&2.do](#)

Robustness check 3

- Program: [Corruption_Final/Do/Robust_3_LPM.do](#)